

CHAPTER-2

2.0 Customer

Introduction

The entities required for the Nykaa E-Commerce System are identified based on the business requirements of an online beauty and personal care platform. Each entity contains attributes, primary keys, foreign keys, and data types to ensure efficient data storage and retrieval. The relationships between entities represent customer registration, product management, brand management, order processing, payment transactions, delivery tracking, and product reviews. This entity analysis provides a strong foundation for designing a normalized relational database while maintaining data integrity and reducing redundancy.

Customer

Attribute	Data Type	Key	Description
Customer_ID	INT	PK	Unique customer ID
First_Name	VARCHAR(50)	-	Customer first name
Last_Name	VARCHAR(50)	-	Customer last name
Gender	VARCHAR(10)	-	Gender
Date_of_Birth	DATE	-	Date of birth
Email	VARCHAR(100)	-	Email address
Phone	VARCHAR(15)	-	Mobile number
Password	VARCHAR(100)	-	Account password
Address	TEXT	-	Customer address
City	VARCHAR(50)	-	City
State	VARCHAR(50)	-	State
Pincode	INT	-	Postal code
Registration_Date	DATE	-	Registration date

2.1 Product

Attribute	Data Type	Key	Description
Product_ID	INT	PK	Unique product ID
Product_Name	VARCHAR(100)	-	Product name
Brand_ID	INT	FK	References Brand
Category_ID	INT	FK	References Category
Price	DECIMAL(10,2)	-	Product price
Stock	INT	-	Available quantity
Description	TEXT	-	Product description



Attribute	Data Type	Key	Description
Skin_Type	VARCHAR(30)	-	Suitable skin type
Rating	DECIMAL(2,1)	-	Product rating
Expiry_Date	DATE	-	Product expiry date

2.2 category

Attribute	Data Type	Key	Description
Category_ID	INT	PK	Unique category ID
Category_Name	VARCHAR(50)	-	Category name
Description	TEXT	-	Category description
Status	VARCHAR(20)	-	Active/Inactive

2.3 Brand

Attribute	Data Type	Key	Description
Brand_ID	INT	PK	Unique brand ID
Brand_Name	VARCHAR(100)	-	Brand name
Country	VARCHAR(50)	-	Country of origin
Contact_Email	VARCHAR(100)	-	Brand email
Phone	VARCHAR(15)	-	Contact number

2.4 Cart

Attribute	Data Type	Key	Description
Cart_ID	INT	PK	Unique cart ID
Customer_ID	INT	FK	References Customer
Product_ID	INT	FK	References Product
Quantity	INT	-	Selected quantity
Added_Date	DATE	-	Date added to cart

2.5 Orders

Attribute	Data Type	Key	Description
Order_ID	INT	PK	Unique order ID
Customer_ID	INT	FK	References Customer
Order_Date	DATE	-	Order date
Total_Amount	DECIMAL(10,2)	-	Total amount
Order_Status	VARCHAR(30)	-	Pending, Shipped, Delivered
Shipping_Address	TEXT	-	Delivery address

2.6 Order Details

Attribute	Data Type	Key Description
Order_Detail_ID	INT	PK Unique order detail ID
Order_ID	INT	FK References Orders
Product_ID	INT	FK References Product
Quantity	INT	- Ordered quantity
Unit_Price	DECIMAL(10,2)	- Product price
Discount	DECIMAL(10,2)	- Discount amount
Total_Price	DECIMAL(10,2)	- Total price

2.7 Payment

Attribute	Data Type	Key Description
Payment_ID	INT	PK Unique payment ID
Order_ID	INT	FK References Orders
Payment_Method	VARCHAR(30)	- UPI, Card, Net Banking, COD
Transaction_ID	VARCHAR(100)	- Transaction reference
Payment_Status	VARCHAR(20)	- Success, Failed, Pending
Payment_Date	DATE	- Payment date
Amount	DECIMAL(10,2)	- Paid amount

2.8 Review

Attribute	Data Type	Key Description
Review_ID	INT	PK Unique review ID
Customer_ID	INT	FK References Customer
Product_ID	INT	FK References Product
Rating	INT	- Rating out of 5
Review_Text	TEXT	- Customer review
Review_Date	DATE	- Review date

2.9 Admin

Attribute	Data Type	Key Description
Admin_ID	INT	PK Unique admin ID
Admin_Name	VARCHAR(100)	- Administrator name

Attribute	Data Type	Key	Description
Email	VARCHAR(100)	-	Email address
Username	VARCHAR(50)	-	Login username
Password	VARCHAR(100)	-	Login password
Role	VARCHAR(30)	-	Admin role

2.10 Relationships

Entity 1	Relationship	Entity 2	Cardinality
Customer	Places	Orders	1 : M
Orders	Contains	Order Details	1 : M
Product	Appears In	Order Details	1 : M
Category	Contains	Product	1 : M
Brand	Manufactures	Product	1 : M
Customer	Adds	Cart	1 : M
Orders	Has	Payment	1 : 1
Customer	Writes	Review	1 : M
Product	Receives	Review	1 : M
Admin	Manages	Product	1 : M
Admin	Manages	Orders	1 : M

